Q-band ART: Adaptive Band Suppression for Anti-Rollover Tracking of Semi-Trailer Tank Trucks

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***Abstract*—Semi-trailer tankers are prone to rollover because lateral acceleration excites suspension roll and liquid sloshing. Existing anti-rollover tracking methods often embed liquid states, roll states, or lateral load transfer ratio predictions in model predictive control, increasing dependence on liquid calibration, trailer-side sensing, and online model dimension. This paper proposes Q-band ART (Anti-Rollover Tracking), an adaptive band-suppression framework that penalizes predicted lateral-acceleration energy in rollover-critical frequency bands. A reduced cart-roll-sloshing model explains why critical bands should be identified from the coupled acceleration-to-risk response, and the Q-band term is embedded into a nonlinear predictive tracking controller with acceleration smoothing, soft band constraints, preview-adaptive weights, extended-state disturbance compensation, and hitch-angle-based parameter scheduling. High-fidelity TruckSim–STAR-CCM+ co-simulations show that Q-band ART prevents rollover in aggressive five-pins maneuvers where conventional preview tracking and acceleration-amplitude limiting fail. Compared with the state-of-the-art model-based active steering method, it achieves comparable anti-rollover tracking while requiring only tractor-side information and a low-order prediction model. Ablation results confirm that frequency-selective excitation suppression, rather than amplitude limiting alone, is the key mechanism. Open-source implementation is available at https://github.com/KomasaQi/HitchOpen-THICV-Stack**

***Index Terms*—Anti-rollover control, semi-trailer tanker, liquid sloshing, path tracking, model predictive control, frequency-band suppression.**

I. Introduction

SEMI-TRAILER tank trucks are indispensable for liquid hazardous-chemical logistics, but partially filled tanks make rollover a severe safety risk. Accident reports show that tank truck rollovers are often triggered by sharp steering, emergency avoidance, curved-road overspeed, or driver error, and may lead to deadly secondary disasters [1]–[3]. Because practical operation spans partial filling rates [4], [5], free-surface sloshing must be considered in autonomous tank truck tracking.

Free-surface excitation generates additional sloshing, and medium filling rates can be less stable than empty or fully filled cases. Tank sloshing has been modeled by quasi-static centroid-shift methods, mechanical equivalent models (MEMs), computational fluid dynamics, and experiments [6]–[10]. Among them, MEMs, especially pendulum models, are widely used because they balance interpretability and real-time computation, although dominant sloshing modes still require accurate calibration [11]–[14].

For semi-trailer tank trucks, rollover prevention is further complicated by tractor-trailer articulation. High-fidelity models may include tractor and trailer lateral, yaw, and roll motions together with liquid states [15]–[18]. Such models are useful offline, but online use is limited by trailer replacement, non-unified trailer interfaces, limited trailer-side sensing, and uncertain liquid/suspension parameters. Active suspension, steering, and differential braking have all been studied for rollover prevention [19]–[21]; tank truck-specific controllers have used PID, LQR, MPC, model-free adaptive control (MFAC), and robust control with MEMs [22]–[27].

Most existing controllers still rely on explicit liquid states, roll states, or lateral load transfer ratio (LTR) predictions, which is not always suitable for autonomous semi-trailer tank trucks with limited trailer sensing and embedded computing resources. Model predictive control (MPC) can combine tracking, actuator, acceleration, and safety constraints [28], but high-dimensional states and dense constraints challenge

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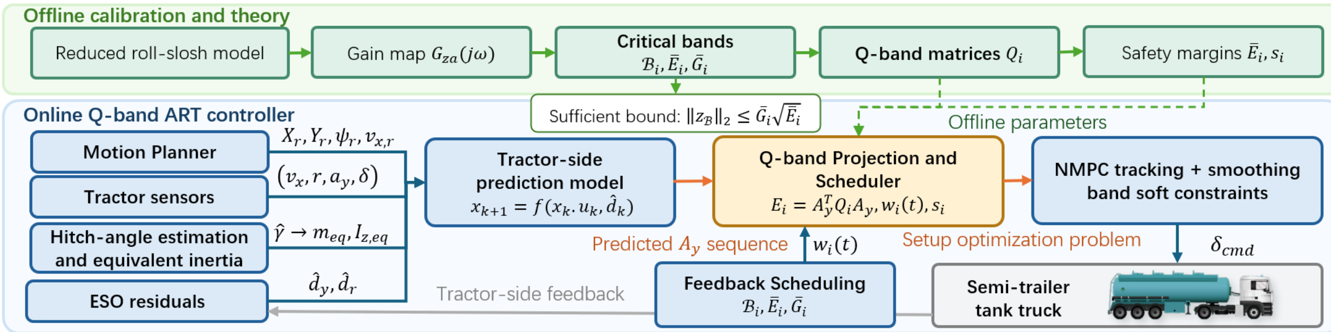

Fig. 1: Architecture of Q-band ART. Offline calibration identifies the critical bands, energy thresholds, Q-band matrices, and conservative margins. The online controller then combines tractor-side prediction, ESO residual compensation, hitch-angle-based equivalent-parameter scheduling, Q-band projection, and NMPC to output steering commands using tractor-side sensing and precomputed Q-band parameters.

real-time implementation [29]. This paper instead suppresses rollover excitation at the input side: the predicted lateral-acceleration sequence is projected onto vehicle-liquid critical bands, and the band energy is penalized and softly constrained. The workflow is shown in Fig. 1.

The contributions are threefold. First, a Q-band anti-excitation view is introduced for semi-trailer tank truck tracking, avoiding mandatory online liquid-state modeling and observation. Second, a multi-point NMPC tracking controller with disturbance compensation and trailer-free modeling is designed for semi-trailers. Third, the Q-band term is integrated predictively for anti-rollover tracking and validated through high-fidelity co-simulation and ablation studies.

II. Q-Band Suppression Theory

A. Reduced Roll-Sloshing Excitation Model

Full liquid-state prediction requires liquid-state observation [14], which is not always available. A reduced model is therefore used to explain why input-side band suppression can reduce rollover risk and how dangerous frequency components are identified. As illustrated in Fig. 2, the unsprung lateral motion is represented by a lateral cart, the sprung mass by a roll pendulum, and the dominant liquid mode by a pendulum attached to the tank body. Taking lateral acceleration a_y as input, the coupled roll-sloshing dynamics can be linearized as

$$\mathbf{M}\ddot{\mathbf{x}}_r + \mathbf{C}\dot{\mathbf{x}}_r + \mathbf{K}\mathbf{x}_r = \mathbf{B}_a a_y, \quad \mathbf{x}_r = [\phi, \theta]^{\mathrm{T}}, \tag{1}$$

where ϕ is the tank-body roll angle, θ is the relative liquid sloshing angle, $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ are the equivalent inertia, damping, and stiffness matrices, and $\mathbf{B}_a$ maps the base lateral acceleration into the roll-slosh coordinates. The off-diagonal terms in $\mathbf{M}$ and $\mathbf{K}$ describe the coupled free mode between suspension roll and liquid motion as

$$\det(\mathbf{K} - \omega^2\mathbf{M}) = 0. \tag{2}$$

Thus, the dangerous frequency is not the isolated liquid or roll natural frequency; the coupled modes generate a broad peak.

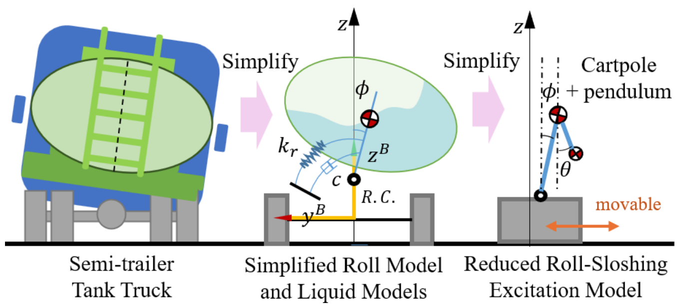

Fig. 2: Reduced lateral model. The online controller only needs the frequency response between input and output.

B. Critical Band and Risk Bound

Let the rollover-risk output be

$$z = C_z\mathbf{x}_r + D_z a_y, \tag{3}$$

where z represents LTR, roll angle, roll rate, or another rollover index, and in this work roll rate or a residual related to roll/yaw amplification is used as the measurable risk-output when available. From (1), the input-output frequency response is

$$G_{za}(s) = C_z(s^2\mathbf{M} + s\mathbf{C} + \mathbf{K})^{-1}\mathbf{B}_a + D_z. \tag{4}$$

The Q-band $\mathcal{B}_i = [f_i^-, f_i^+]$ is selected around the broad-peak region of $|G_{za}(j2\pi f)|$, where f_i^- and f_i^+ are the lower and upper frequencies of the ith band. Here, dangerous bands are calibrated offline from sweep analysis and high-fidelity TruckSim–STAR-CCM+ co-simulation. The primary band is 0.35–0.75 Hz, with secondary coverage of 0.75–1.45 Hz for higher coupled roll-sloshing excitations.

Safety could be stated as a sufficient frequency-domain condition. Let $a_{y,\mathcal{B}}$ denote the lateral-acceleration component in a target band $\mathcal{B}$, and let $z_{\mathcal{B}}$ be the corresponding risk-output component. Around the calibrated operating point, assume that the linearized system is stable and

$$|G_{za}(j2\pi f)| \le \bar{G}_{\mathcal{B}}, \qquad f \in \mathcal{B}. \tag{5}$$

Then Parseval's relation gives

$$\|z_{\mathcal{B}}\|_2^2 \le \bar{G}_{\mathcal{B}}^2 \|a_{y,\mathcal{B}}\|_2^2. \tag{6}$$

Thus, if $\|a_{y,\mathcal{B}}\|_2^2 \le \bar{E}_{\mathcal{B}}$, where $\bar{E}_{\mathcal{B}}$ is the given band-energy threshold,

$$\|z_{\mathcal{B}}\|_2 \le \bar{G}_{\mathcal{B}} \sqrt{\bar{E}_{\mathcal{B}}}. \tag{7}$$

Considering static load-transfer contribution and bounded residual terms, a conservative rollover margin can be written as

$$|\mathrm{LTR}_{\mathrm{stat}}| + \bar{G}_{\mathcal{B}} \sqrt{\bar{E}_{\mathcal{B}}} + R_{\perp} + R_0 + R_m \le \mathrm{LTR}_{\mathrm{lim}}, \tag{8}$$

where $R_{\perp}$, R_0, and R_m bound non-target-band excitation, initial roll/slosh energy, and model or sensing errors, respectively.

Proof: Inequality (6) follows from the frequency-domain gain bound of the stable linearized input-output model. Substituting the band-energy threshold gives (7). Adding the bounded static, off-band, initial-condition, and model-error terms yields (8). The condition is sufficient rather than necessary, with conservatism from linearization, band discretization, gain upper bounds, and residual margins.

This proof explains why limiting the band energy of a_y can reduce rollover-risk excitation even when liquid phase and true wheel loads are not observed online. The discrete Q-band matrix used by the controller is an implementable approximation of $\|a_{y,\mathcal{B}}\|_2^2$ over the prediction horizon.

III. Adaptive Q-Band Suppression Controller

A. Tractor Prediction Model and ESO

The online optimizer uses a low-dimensional tractor-side nonlinear prediction model. Trailer force, liquid sloshing, tire mismatch, road slope, and parameter errors are not explicit states; they are collected as residual terms. Let v_x, v_y, r, and δ denote tractor longitudinal speed, lateral speed, yaw rate, and front-wheel steering angle. With front and rear cornering stiffnesses C_f and C_r, the tire slip angles and lateral forces are

$$\alpha_f = \delta - \frac{v_y + l_f r}{v_x}, \qquad \alpha_r = -\frac{v_y - l_r r}{v_x}, \tag{9}$$

$$F_{yf} = C_f \alpha_f, \qquad F_{yr} = C_r \alpha_r, \tag{10}$$

where l_f and l_r are the distances from tractor center of gravity to the front and rear axles. The nominal prediction model is

$$\begin{aligned} \dot{v}_y &= \frac{F_{yf}\cos\delta + F_{yr}}{m_{\mathrm{eq}}} - v_x r + d_y, \\ \dot{r} &= \frac{l_f F_{yf}\cos\delta - l_r F_{yr}}{I_{z,\mathrm{eq}}} + d_r, \end{aligned} \tag{11}$$

where m_{eq} and $I_{z,\mathrm{eq}}$ are the hitch-angle-scheduled equivalent lateral mass and yaw inertia, and d_y, d_r are lumped lateral and yaw residuals. The lateral acceleration used by Q-band is computed from the prediction model as

$$a_{y,k}^{\mathrm{pred}} = \dot{v}_{y,k} + v_{x,k} r_k. \tag{12}$$

An extended-state observer (ESO) estimates the yaw-channel residual:

$$\begin{aligned} \dot{\hat{r}} &= \frac{l_f \hat{F}_{yf}\cos\delta - l_r \hat{F}_{yr}}{I_{z,\mathrm{eq}}} + \hat{d}_r + \beta_{r1}(r - \hat{r}), \\ \dot{\hat{d}}_r &= \beta_{r2}(r - \hat{r}), \end{aligned} \tag{13}$$

where $\hat{r}$ and $\hat{d}_r$ are the estimated yaw rate and yaw residual, and $\beta_{r1}, \beta_{r2} > 0$ are observer gains. The lateral residual is obtained from the difference between the measured lateral acceleration and the kinematic centripetal component:

$$\hat{d}_y = a_y^{\mathrm{centri}} - a_y^{\mathrm{meas}} = v_x r - (a_y^{\mathrm{imu}} - \Delta a_{y,0}), \tag{14}$$

where a_y^{imu} is the IMU lateral acceleration, $\Delta a_{y,0}$ is its slowly updated zero offset, and $a_y^{\mathrm{centri}} = v_x r$ is the low-dimensional centripetal estimate. In the prediction horizon, the residual is held or decayed as

$$\hat{d}_{\ell|k} = \lambda_d^{\ell} \hat{d}_k, \qquad 0 < \lambda_d \le 1, \tag{15}$$

where ℓ is the prediction step and λ_d is the residual forgetting factor.

B. Hitch-Angle and Equivalent Inertia

The equivalent parameters in (11) use a tractor-trailer articulation prior. Let the tractor and trailer headings be ψ_1 and ψ_2, and define the hitch angle as

$$\gamma = \psi_1 - \psi_2. \tag{16}$$

With measured tractor yaw rate r_m, the trailer heading and hitch angle are estimated by

$$\begin{aligned} \dot{\hat{\psi}}_2 &= K_v \frac{v_x}{L_2}\left(\frac{L_h}{L_1}\tan\delta\cos\hat{\gamma} + \sin\hat{\gamma}\right), \\ \dot{\hat{\gamma}} &= r_m - \dot{\hat{\psi}}_2. \end{aligned} \tag{17}$$

Here L_1 is the tractor wheelbase, L_2 is the distance from the hitch to the trailer axle group, and L_h is the distance from the tractor rear axle to the hitch. The correction factor

$$K_v = 1 - \frac{m_2 v_x^2 e}{L_1 L_2 k_3} \tag{18}$$

approximates trailer-axle sideslip, where m_2 is trailer mass, e is the equivalent hitch-to-trailer geometric offset, and k_3 is the trailer axle cornering stiffness. When these trailer tire parameters are not available, $K_v = 1$ is used.

The trailer contribution is projected into tractor lateral inertia and yaw inertia:

$$m_{x,\mathrm{eq}}(\hat{\gamma}) = m_1 + m_2\cos^2\hat{\gamma}, \qquad m_{y,\mathrm{eq}}(\hat{\gamma}) = m_1 + m_2\sin^2\hat{\gamma}, \tag{19}$$

$$I_{z,\mathrm{eq}}(\hat{\gamma}) = I_{z1} + I_{z2}\left(\frac{d_1}{d_2}\right)^2 \cos^2\hat{\gamma} + m_2 d_1^2 \sin^2\hat{\gamma}. \tag{20}$$

Here m_1 is tractor mass, I_{z1} and I_{z2} are tractor and trailer yaw inertias, and d_1, d_2 are the equivalent geometric distances used to project trailer yaw inertia to the tractor yaw axis. In the NMPC model, $m_{\mathrm{eq}} = m_{y,\mathrm{eq}}(\hat{\gamma})$, and $I_{z,\mathrm{eq}}$ is evaluated from (20), after low-pass filtering, upper/lower saturation, and rate limiting.

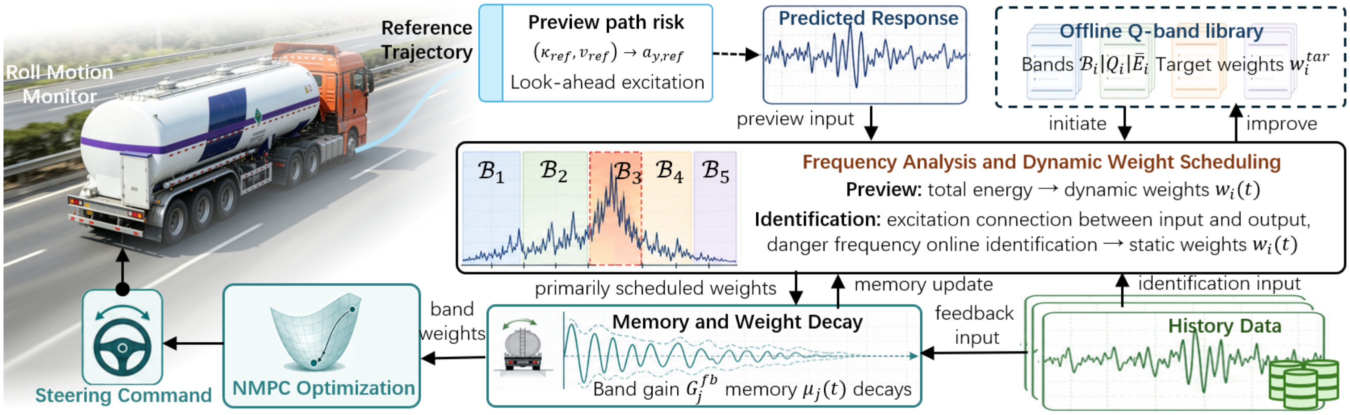

Fig. 3: Adaptive Q-band scheduling. Preview risk raises the band weights before aggressive path segments, while feedback monitoring strengthens dynamically amplified bands and delays weight release after residual sloshing.

C. Band-Energy Projection

Let the prediction horizon contain N samples with sampling time Δt. The predicted lateral-acceleration sequence is

$$\mathbf{A}_y = [a_{y,0}, a_{y,1}, \ldots, a_{y,N-1}]^{\mathrm{T}}, \tag{21}$$

where $a_{y,k}$ is computed from (12) for a dynamic predictor, or from $a_{y,k} \approx v_{x,k}^2 \kappa_k$ for a kinematic predictor with path curvature κ_k. For a critical band $\mathcal{B}_i = [f_i^-, f_i^+]$, choose discrete frequencies f_{ij} and time grid $t_k = k\Delta t$. Define

$$\mathbf{c}_{ij} = [\cos(2\pi f_{ij} t_0), \ldots, \cos(2\pi f_{ij} t_{N-1})]^{\mathrm{T}}, \tag{22}$$

$$\mathbf{s}_{ij} = [\sin(2\pi f_{ij} t_0), \ldots, \sin(2\pi f_{ij} t_{N-1})]^{\mathrm{T}}. \tag{23}$$

The predicted energy in the ith band is

$$E_i(\mathbf{A}_y) = \mathbf{A}_y^{\mathrm{T}} \mathbf{Q}_i \mathbf{A}_y, \tag{24}$$

$$\mathbf{Q}_i = \frac{1}{N^2} \sum_j (\mathbf{c}_{ij}\mathbf{c}_{ij}^{\mathrm{T}} + \mathbf{s}_{ij}\mathbf{s}_{ij}^{\mathrm{T}}) \succeq 0. \tag{25}$$

The matrix $\mathbf{Q}_i$ is precomputed from Δt, N, and $\mathcal{B}_i$. It adds no vehicle state and only projects $\mathbf{A}_y$ onto dangerous frequency components.

D. Optimization Problem

Let $\mathbf{x}_k$ be the prediction state, $u_k = \delta_{\mathrm{cmd},k}$ the steering command, $\mathbf{U} = [u_0, \ldots, u_{N-1}]^{\mathrm{T}}$, and $\mathbf{S} = [s_1, \ldots, s_{n_b}]^{\mathrm{T}}$ the nonnegative slack variables for n_b Q-bands. The Q-band ART objective is

$$\begin{aligned} \min_{\mathbf{U},\mathbf{S}} \; & J_{\mathrm{trk}} + J_u + J_{\Delta u} + w_a \|\mathbf{A}_y\|_2^2 + w_1 \|\mathbf{D}_1 \mathbf{A}_y\|_2^2 \\ & + w_2 \|\mathbf{D}_2 \mathbf{A}_y\|_2^2 + \sum_{i=1}^{n_b} w_i(t) \mathbf{A}_y^{\mathrm{T}} \mathbf{Q}_i \mathbf{A}_y + \sum_{i=1}^{n_b} \rho_i s_i^2. \end{aligned} \tag{26}$$

Here $J_u = w_u \|\mathbf{U}\|_2^2$, $J_{\Delta u} = w_{\Delta u} \|\mathbf{D}_u \mathbf{U}\|_2^2$, and w_a, w_1, w_2, ρ_i are positive tuning weights. The first- and second-order difference matrices are calculated by

$$(\mathbf{D}_1 \mathbf{A}_y)_k = \frac{a_{y,k+1} - a_{y,k}}{\Delta t}, \tag{27}$$

$$(\mathbf{D}_2 \mathbf{A}_y)_k = \frac{a_{y,k+2} - 2a_{y,k+1} + a_{y,k}}{\Delta t^2}. \tag{28}$$

The tracking cost uses the previewed reference position and heading:

$$J_{\mathrm{trk}} = w_{xy}(\|\mathbf{X} - \mathbf{X}_{\mathrm{ref}}\|_2^2 + \|\mathbf{Y} - \mathbf{Y}_{\mathrm{ref}}\|_2^2) + w_\psi \|\boldsymbol{\psi} - \boldsymbol{\psi}_{\mathrm{ref}}\|_2^2, \tag{29}$$

where $\mathbf{X}$, $\mathbf{Y}$, and $\boldsymbol{\psi}$ collect predicted tractor positions and headings over the horizon, and the subscript "ref" denotes the path-preview sequence.

The constraints are

$$\begin{aligned} \mathbf{x}_{k+1} &= f(\mathbf{x}_k, u_k, \hat{d}_{k|k}, m_{\mathrm{eq}}, I_{z,\mathrm{eq}}), \\ |a_{y,k}| &\le a_{y,\max}^{\mathrm{dyn}}, \\ u_{\min} &\le u_k \le u_{\max}, \quad |\Delta u_k| \le \Delta u_{\max}, \\ E_i(\mathbf{A}_y) &\le \bar{E}_i + s_i, \quad s_i \ge 0. \end{aligned} \tag{30}$$

The bound $a_{y,\max}^{\mathrm{dyn}}$ is the speed- and risk-dependent lateral-acceleration limit, and $\bar{E}_i$ is the calibrated safe energy threshold of band $\mathcal{B}_i$.

E. Preview-Adaptive Scheduling

Q-band ART uses preview risk as the main trigger, and the complete weight-scheduling logic is summarized in Fig. 3. The reference lateral-acceleration sequence $\mathbf{A}_{y,\mathrm{ref}}$ is computed from future path curvature as

$$a_{y,\mathrm{ref},k} = v_{x,k}^2 \kappa_{\mathrm{ref},k}, \qquad \kappa_{\mathrm{ref},k} \approx \frac{\mathrm{wrap}(\psi_{\mathrm{ref},k+1} - \psi_{\mathrm{ref},k})}{\Delta s_k}, \tag{31}$$

where Δs_k is the preview arc-length interval and $\mathrm{wrap}(\cdot)$ keeps heading differences within $[-\pi, \pi]$. The preview risk in band i is

$$R_i^{\mathrm{pre}} = \left[\frac{\mathbf{A}_{y,\mathrm{ref}}^{\mathrm{T}} \mathbf{Q}_i \mathbf{A}_{y,\mathrm{ref}}}{\bar{E}_{i,\mathrm{ref}}} - 1 \right]_+, \tag{32}$$

where $[x]_+ = \max(x, 0)$ and $\bar{E}_{i,\mathrm{ref}}$ is the preview activation threshold. The preview-only activation is

$$\lambda_i^{\mathrm{pre}} = 1 - \exp(-\kappa_{\mathrm{pre}} R_i^{\mathrm{pre}}), \tag{33}$$

where $\kappa_{\rm pre} > 0$ controls how quickly the weight rises before an aggressive maneuver.

F. Feedback Band Monitoring and Residual Sloshing Memory

Preview alone cannot identify bands that become dangerous due to loading, fill level, tire state, or unmodeled trailer dynamics. A feedback monitor is therefore added to detect bands with non-negligible input and amplified measured output. Over a sliding window of M samples, define

$$\mathbf{a}_M(t) = [a_y^{\rm mon}(t-M+1), \dots, a_y^{\rm mon}(t)]^{\rm T}, \quad (34)$$

where $a_y^{\rm mon}$ is the measured or predicted lateral acceleration after offset correction. The monitored output is

$$y^{\rm mon} \in \{r - \hat{r},\ a_y^{\rm imu} - a_y^{\rm pred},\ \dot{\phi}\}, \quad (35)$$

with the output window

$$\mathbf{y}_M(t) = [y^{\rm mon}(t-M+1), \dots, y^{\rm mon}(t)]^{\rm T}. \quad (36)$$

The yaw and acceleration residuals are available from tractor-side sensing, and roll rate $\dot{\phi}$ is used when a roll sensor is available. For a monitoring band $\mathcal{C}_j$, construct $\mathbf{Q}_j^M$ as $\mathbf{Q}_i$ but with window length M. The input and output band energies are

$$E_{a,j}^{\rm fb} = \mathbf{a}_M^{\rm T}\mathbf{Q}_j^M\mathbf{a}_M, \qquad E_{y,j}^{\rm fb} = \mathbf{y}_M^{\rm T}\mathbf{Q}_j^M\mathbf{y}_M, \quad (37)$$

and the observed amplification is

$$G_j^{\rm fb} = \sqrt{\frac{E_{y,j}^{\rm fb}}{E_{a,j}^{\rm fb} + \epsilon}}, \quad (38)$$

where ϵ avoids division by zero. A dynamically dangerous band must satisfy both non-negligible input and amplified output:

$$R_j^{\rm fb} = \left[\frac{E_{a,j}^{\rm fb}}{\bar{E}_{a,j}} - 1\right]_+ \left[\frac{G_j^{\rm fb}}{\bar{G}_j^{\rm fb}} - 1\right]_+. \quad (39)$$

Here $\bar{E}_{a,j}$ is the minimum input-energy threshold, and $\bar{G}_j^{\rm fb}$ is the nominal amplification level from calibration or low-risk data.

To retain residual sloshing after steering input decreases, the feedback risk is stored as

$$\mu_j(t) = \max\{\lambda_m\mu_j(t-\Delta t),\ R_j^{\rm fb}(t)\}, \qquad 0 < \lambda_m < 1, \quad (40)$$

where λ_m is the memory decay factor. If the monitoring bands $\mathcal{C}_j$ are finer than the default controller bands $\mathcal{B}_i$, the feedback memory is mapped to the controller band by the overlap ratio

$$\mu_i^{\rm dyn}(t) = \max_j \sigma_{ij}\mu_j(t), \qquad \sigma_{ij} = \frac{|\mathcal{B}_i \cap \mathcal{C}_j|}{|\mathcal{B}_i|}. \quad (41)$$

Thus, bands observed to be both excited and amplified increase the default Q-band weights.

The preview and feedback activations are fused before filtering:

$$\lambda_i(t) = 1 - \exp[-\kappa_{\rm pre}R_i^{\rm pre} - \kappa_{\rm fb}\mu_i^{\rm dyn}(t)], \quad (42)$$

$$w_{i,\rm tar}^{\rm dyn}(t) = \min\{w_i^{\max},\ w_i^{\rm tar}(1 + \eta_{\rm fb}\mu_i^{\rm dyn}(t))\}, \quad (43)$$

$$w_i^{\rm raw}(t) = w_i^{\min} + \lambda_i(t)\left(w_{i,\rm tar}^{\rm dyn}(t) - w_i^{\min}\right). \quad (44)$$

Here $\kappa_{\rm fb}$ and $\eta_{\rm fb}$ tune the feedback contribution, $w_i^{\min}$ is the low-risk weight, $w_i^{\rm tar}$ is the nominal calibrated target weight, and $w_i^{\max}$ prevents excessive conservatism. Finally, a non-symmetric filter is applied:

$$w_i(t) = (1-\alpha_i)w_i(t-\Delta t) + \alpha_i w_i^{\rm raw}(t), \quad (45)$$

where $\alpha_i = \alpha_\uparrow$ when $w_i^{\rm raw} > w_i(t-\Delta t)$ and $\alpha_i = \alpha_\downarrow$ otherwise, with $\alpha_\uparrow > \alpha_\downarrow$. This makes the weight rise quickly when preview or feedback detects risk, and decay slowly while residual sloshing memory remains.

IV. Validation

A. Co-Simulation Setup

Validation uses the TruckSim–STAR-CCM+ vehicle–liquid coupling platform reported in [27]. The vehicle is the default full-load 49-ton 3A sleeper tractor with trailer. The tank is filled to 50%, and the liquid–vehicle interaction is generated by a three-dimensional STAR-CCM+ sloshing model coupled with the nonlinear TruckSim vehicle model. This high-fidelity platform is used only for validation; the Q-band controller does not use liquid states, trailer-side states, or real-time fluid forces as feedback.

The main comparison uses the continuous five-pins maneuver sketched in Fig. 4. Speed increases linearly from 60 km/h to 100 km/h within 60 s. The five double-lane-change (DLC) maneuvers become progressively more aggressive, with lane-change distances of 100, 81.25, 62.5, 43.75, and 25 m for DLC1–DLC5, respectively.

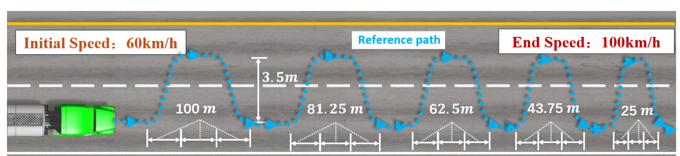

Fig. 4: Continuous five-pins maneuver: speed ramps from 60 km/h to 100 km/h, and the five DLC distances are 100, 81.25, 62.5, 43.75, and 25 m.

Three controllers are compared. TruckSim preview tracking (PT) tracks the path without rollover awareness. The full-state active rollover tracking controller (Full-ART) from [27] explicitly uses trailer and liquid dynamics with liquid-state observation and trailer-state feedback. It can approach the physical limit but has high model, calibration, and sensing requirements. Q-band ART instead uses tractor-side motion information and suppresses dangerous lateral-excitation bands. Evaluation uses high-fidelity LTR, path error, speed, steering angle, and average solve time; LTR is used only as an offline safety metric, not as an online controller state.

B. Continuous Five-Pins Results

Figure 5 shows the continuous five-pins comparison. PT follows the first pins tightly but keeps pursuing the geometric path after the maneuver becomes dynamically unsafe. Its LTR

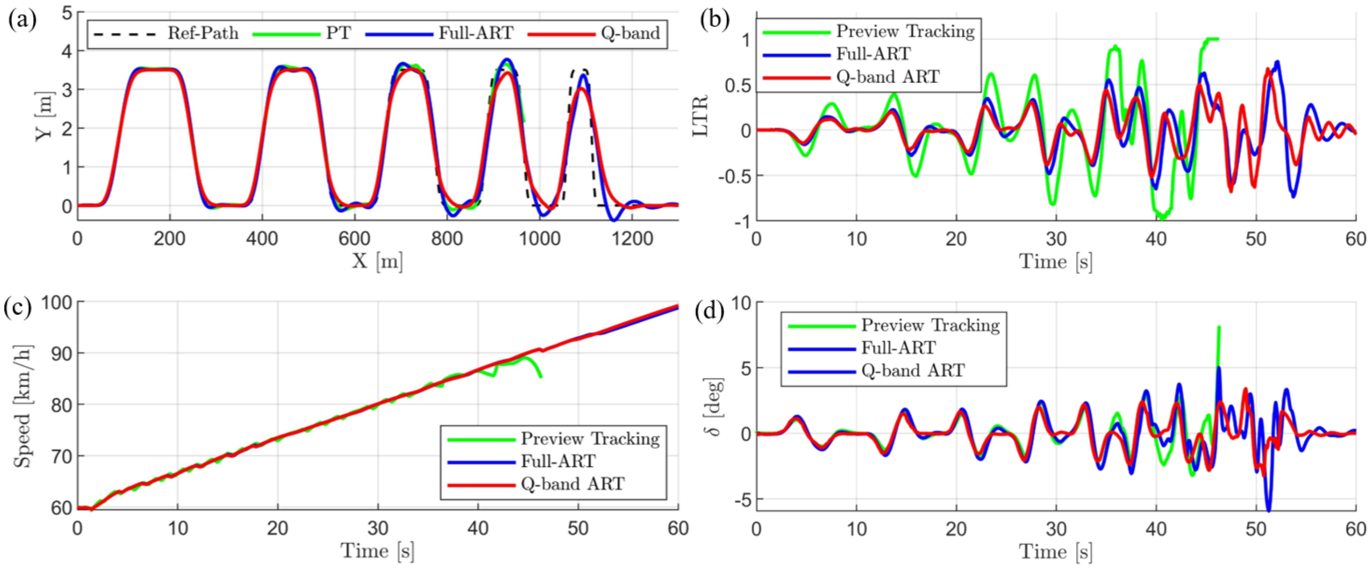

Fig. 5: Continuous five-pins comparison: (a) trajectory tracking, (b) lateral load transfer ratio, (c) longitudinal speed, and (d) front steering angle. PT denotes the TruckSim preview tracking baseline, Full-ART denotes the full-state active rollover tracking controller in [27], and Q-band ART denotes the proposed controller.

increases rapidly in later pins and reaches the rollover boundary, confirming that pure preview tracking is inadequate for a high-speed liquid tank semi-trailer even when the reference path is smooth. The failure is not caused by an isolated steering spike; it appears after repeated lane changes, where the path input keeps injecting lateral-acceleration energy into the coupled roll-sloshing response.

Full-ART and Q-band ART both complete the maneuver without rollover. Full-ART tracks slightly tighter in the most aggressive pins because it uses richer vehicle–liquid prediction and full-state feedback. Q-band ART intentionally relaxes the path in high-risk portions, especially near the fifth pin, preventing lateral excitation from feeding dominant roll–sloshing bands. Its trajectory is more conservative, but its LTR response remains comparable to Full-ART, with nearly identical speed history and practical steering commands.

These results highlight the main tradeoff: Full-ART approaches the limit more aggressively through detailed trailer–liquid modeling, whereas Q-band ART obtains a similar rollover-prevention effect with tractor-side measurements and a low-order prediction model. The comparison is therefore not intended to show that Q-band ART dominates a full-state controller near the physical limit; rather, it shows that most of the rollover-prevention benefit can be recovered by shaping the excitation spectrum before detailed liquid or trailer states are needed online.

The four panels in Fig. 5 should be read jointly. The speed profiles are nearly identical, so the safety difference is not produced by slowing down the vehicle. The steering histories also remain within practical amplitudes, indicating that Q-band ART does not rely on abrupt steering intervention. Instead, the trajectory panel shows a selective relaxation of path tracking when the reference path becomes increasingly aggressive. This relaxation changes the timing and spectral content of lateral acceleration, which is reflected by the reduced LTR build-up in the later pins. In other words, the controller trades a small path deviation for a much larger reduction in excitation delivered to the coupled roll–sloshing dynamics.

C. Q-Band Ablation Study

The ablation study uses five independent DLC simulations because, once rollover occurs in an earlier pin, later pins are no longer physically meaningful. Each DLC starts from the corresponding rollover-free Q-band condition. The variants include Base, AyLim, AyCost, dAy, ddAy, Smooth, Qband, Qband-Soft, Fixed-Q, Fb-Q, Pre-Q, and soft-constrained combinations.

The ablation design separates three effects that are often mixed in MPC safety tuning. First, Base represents geometric tracking alone. Second, AyLim, AyCost, dAy, ddAy, and Smooth test whether amplitude and smoothness regulation of a_y is sufficient. Third, the Qband variants test whether penalizing frequency-selective energy provides an additional mechanism. For rollover cases, the reported peak LTR is clipped at the physical boundary in Table I; the dagger indicates that the actual simulation reached rollover rather than that the plotted signal was numerically saturated.

Figure 6 and Table I summarize the ablation results. In DLC1 and DLC2, all variants remain far from the rollover boundary with small tracking errors, showing that Q-band terms do not dominate low-risk maneuvers. As the lane-change distance decreases in DLC3, Q-band variants reduce peak $|\mathrm{LTR}_{\mathrm{est}}|$ while keeping RMS tracking error acceptable. The table also shows why path error alone is a misleading objective: Base has the best RMS and peak tracking errors,

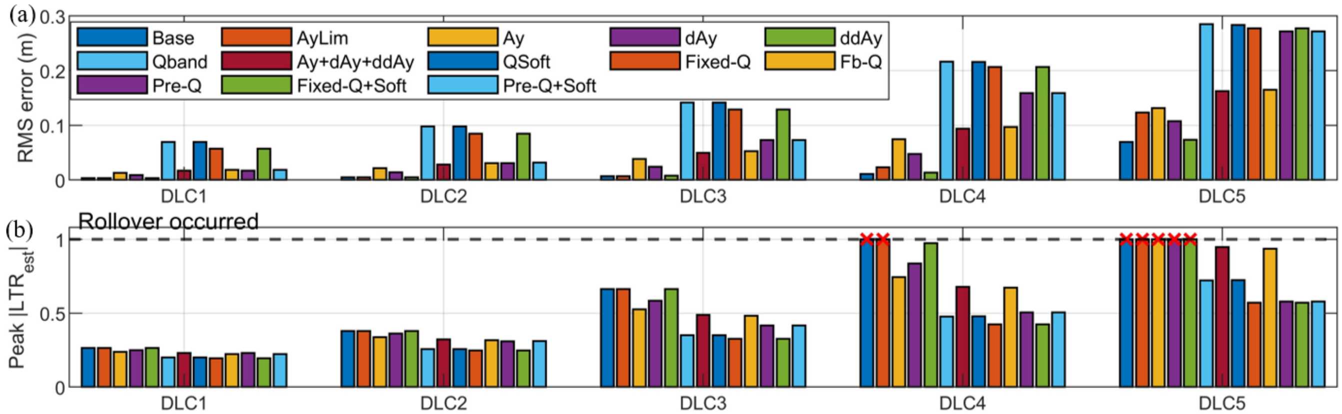

Fig. 6: Ablation study of the Q-band mechanism in five independent DLC segments: (a) RMS tracking error and (b) peak estimated LTR. Red crosses at the dashed $|\mathrm{LTR}_{\mathrm{est}}| = 1$ line indicate simulations in which rollover occurred, not numerical clipping.

TABLE I: Ablation metrics

Var.	pk $\lvert\mathrm{LTR}\rvert$	rms LTR	rms e_y	pk $\lvert e_y\rvert$	t_s
Base	1.0000†	0.3021	**0.0266**	**0.2873**	**12.6**
AyLim	1.0000†	0.2912	0.0474	0.4148	14.7
AyCost	1.0000†	0.2545	0.0602	0.3689	13.5
dAy	1.0000†	0.2756	0.0460	0.3540	**13.0**
ddAy	1.0000†	0.2995	**0.0284**	**0.3077**	17.7
Qband	0.7210	**0.1811**	0.1582	0.8189	22.6
Smooth	0.9474	0.2357	0.0753	0.4609	21.2
Qband-Soft	0.7244	0.1815	0.1578	0.8144	32.8
Fixed-Q	**0.5719**	**0.1687**	0.1494	0.7740	23.5
Fb-Q	0.9369	0.2333	0.0772	0.4650	26.0
Pre-Q	0.5791	0.1880	0.1232	0.7738	22.3
Fixed-Q+Soft	**0.5719**	**0.1687**	0.1494	0.7740	30.6
Pre-Q+Soft	**0.5790**	0.1877	0.1233	0.7738	30.6

Note: e_y is in m and t_s is in ms. Lower is better; bold/bold-underlined values mark the best/second-best. For rollover cases, pk $|\mathrm{LTR}|$ is reported at the physical boundary 1.0000 and marked by †.

yet it is unsafe. Safe tracking here requires deliberately giving up a small amount of geometric accuracy when the reference path excites the tank-trailer dynamics.

The difference becomes decisive in DLC4 and DLC5. Base and several acceleration-only variants reach the rollover boundary. AyLim still rolls over despite the same lateral-acceleration amplitude limit, proving that rollover prevention cannot be explained by limiting only peak a_y. Even with bounded acceleration magnitude, repeated excitation near dominant roll–sloshing bands can accumulate energy and trigger rollover. Smooth acceleration penalties reduce this tendency but remain indirect: they suppress rapid changes over the whole spectrum, whereas Q-band weighting targets the frequency range where lateral acceleration is most efficiently converted into rollover risk.

This also explains why the ablation is reported with independent DLC segments rather than a single failed continuous run. If a controller rolls over in an early severe segment, all subsequent path-error statistics would be dominated by post-rollover motion and would no longer describe tracking-control quality. Resetting each segment from the corresponding rollover-free Q-band state keeps the comparison focused on the local controller decision: whether the same previewed path is followed tightly, softened through generic acceleration penalties, or reshaped specifically to avoid the dangerous band. The resulting contrast is sharper than a survival-only score and directly supports the frequency-domain argument in Section II.

Among Q-band variants, fixed Q-band weighting already reduces peak LTR in severe DLCs, showing that frequency-selective penalization is the core contributor. Qband alone lowers rollover risk substantially, but its larger tracking error indicates that a single strong band penalty tends to be conservative. Fixed-Q and Fixed-Q+Soft give the lowest LTR values because the dangerous bands are known for this 50% filling condition; Pre-Q and Pre-Q+Soft trade a small amount of peak LTR margin for lower tracking error by activating the penalty mainly when the previewed path becomes risky. This behavior is desirable for deployment because it avoids over-penalizing low-risk tracking while still reacting before the vehicle enters the most aggressive lane change.

Feedback adaptation is interpreted as a supplementary layer rather than the primary trigger. Since it requires measured amplification, it reacts after part of the risky response has appeared; therefore, it is less effective than preview or fixed band knowledge in the most severe open-loop excitation. Its value is different: if the actual vehicle has a shifted dangerous band due to fill level, tire state, or trailer replacement, feedback memory can delay weight release and prevent repeated excitation from being treated as a new low-risk maneuver. Soft constraints play a related role at the optimizer level by allowing controlled tracking relaxation instead of forcing an infeasible compromise between path error and rollover margin.

The solve-time column also supports the intended design logic. Adding Q-band terms increases computation from the tracking-only case, but all variants remain in the same order of magnitude because the band projection is a precomputed quadratic form in the predicted acceleration sequence. The controller therefore gains frequency-selective rollover awareness without introducing liquid states, wheel-load states, or a high-dimensional trailer–fluid prediction model. Overall, the ablation confirms that suppressing lateral excitation energy in dangerous bands is the main protective mechanism, while preview scheduling, feedback memory, and soft constraints determine how early and smoothly this protection is activated.

V. Conclusion

This paper proposed Q-band ART, an adaptive frequency-band suppression method for anti-rollover tracking of semi-trailer tank trucks. Instead of using liquid sloshing states or wheel-load-based LTR as online controller states, it suppresses predicted lateral-acceleration energy in critical vehicle-liquid bands. A reduced roll-sloshing model provides the frequency-domain safety interpretation, and the Q-band term is integrated with acceleration smoothing, soft constraints, preview-adaptive scheduling, ESO compensation, and hitch-angle-based parameter scheduling.

Continuous five-pins co-simulation and independent-DLC ablation show that Q-band ART avoids rollover where pure preview tracking and acceleration-amplitude limiting fail. Compared with full-state active steering, it provides comparable anti-rollover tracking with lower dependence on liquid-state observation, trailer-side feedback, and high-dimensional online models. The results indicate that input-side critical-band suppression is a practical complement to explicit rollover-state constraints. Future work will refine critical-band calibration across filling rates and integrate Q-band scheduling with combined steering and braking control.

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